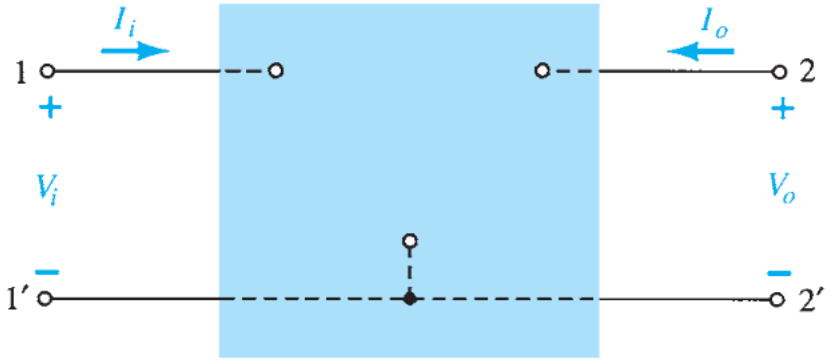
# Hybrid Equivalent Model and Feedback Amplifiers

Ginanjar S. Adi

# THE HYBRID EQUIVALENT MODEL (Review)

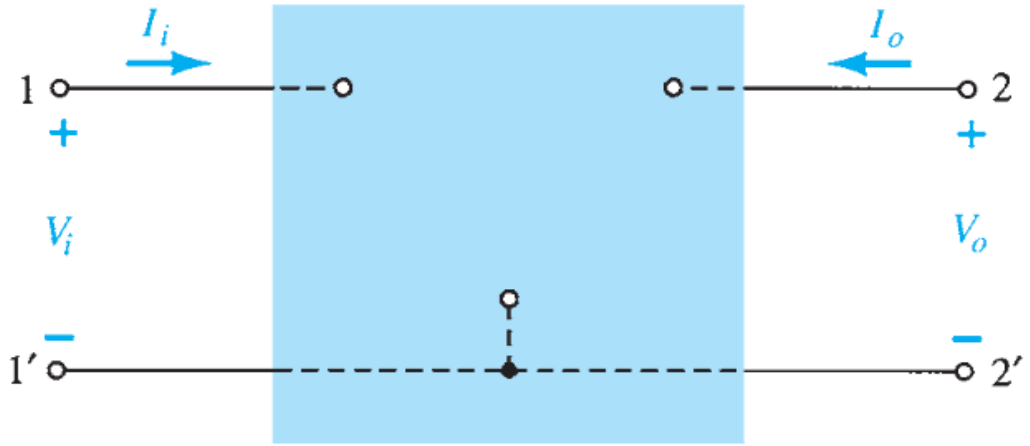
***The parameters of the hybrid equivalent circuit are defined in general terms for any operating conditions.***

- In other words, the hybrid parameters may not reflect the actual operating conditions but simply provide an indication of the level of each parameter to expect for general use.
- The description of the hybrid equivalent model will begin with the general two-port system of figure bellow:



# THE HYBRID EQUIVALENT MODEL (Cont…)

The parameters relating the four variables are called *h-parameters*, from the word "hybrid." The term *hybrid* was chosen because the mixture of variables (*V* and *I*) in each equation results in a "hybrid" set of units of measurement for the *h*-parameters.




$$V_i = h_{11}I_i + h_{12}V_o$$

$$I_o = h_{21}I_i + h_{22}V_o$$

# THE HYBRID EQUIVALENT MODEL (Cont…)

$$V_i = h_{11}I_i + h_{12}V_o$$

- **$h_{11}$** If we arbitrarily set $V_o$ = 0 (short circuit the output terminals) and solve for $h_{11}$, we find (*short-circuit input-impedance parameter*)

$$h_{11} = \left.\frac{V_i}{I_i}\right|_{V_o=0} \quad \text{ohms}$$

- **$h_{12}$** If $I_i$ is set equal to zero by opening the input leads, the following results for $h_{12}$ (*open-circuit reverse transfer voltage ratio parameter*)

$$h_{12} = \left.\frac{V_i}{V_o}\right|_{I_i=0} \quad \text{unitless}$$

# THE HYBRID EQUIVALENT MODEL (Cont…)

$$I_o = h_{21}I_i + h_{22}V_o$$

- ***$h_{21}$*** If $V_o$ is set equal to zero by again shorting the output terminals, the following results for $h_{21}$ (*short- circuit forward transfer current ratio parameter*)

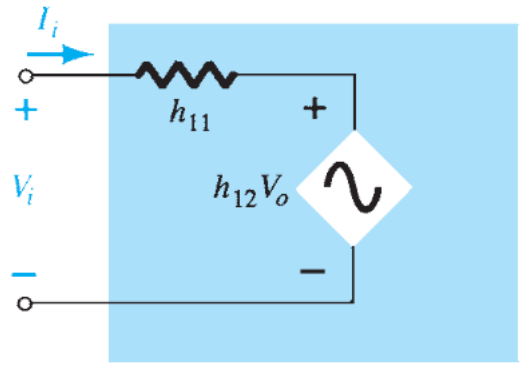
$$h_{21} = \left.\frac{I_o}{I_i}\right|_{V_o=0}$$ unitless

- ***$h_{22}$*** can be found by again opening the input leads to set $I_1 = 0$ and solving for $h_{22}$ (*open-circuit output admittance parameter*)

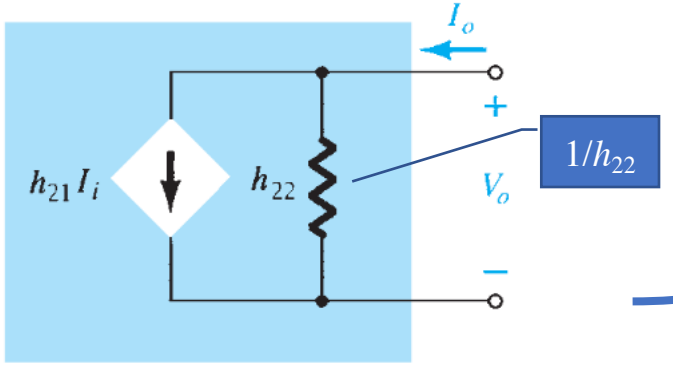
$$h_{22} = \left.\frac{I_o}{V_o}\right|_{I_i=0}$$ siemens

# THE HYBRID EQUIVALENT MODEL (Cont…)

- KVL (Input)




- KCL (Output)

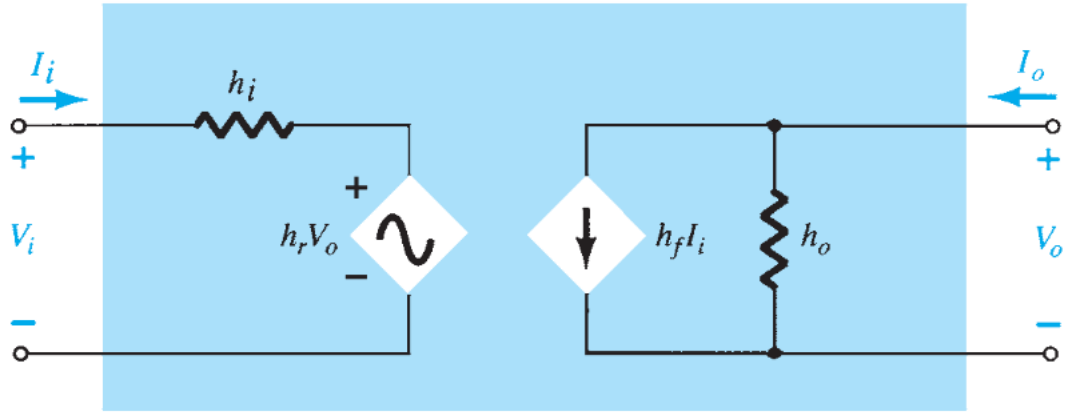



$h_{11} \rightarrow$ *i*nput resistance $\rightarrow h_i$

$h_{12} \rightarrow$ *r*everse transfer voltage ratio $\rightarrow h_r$

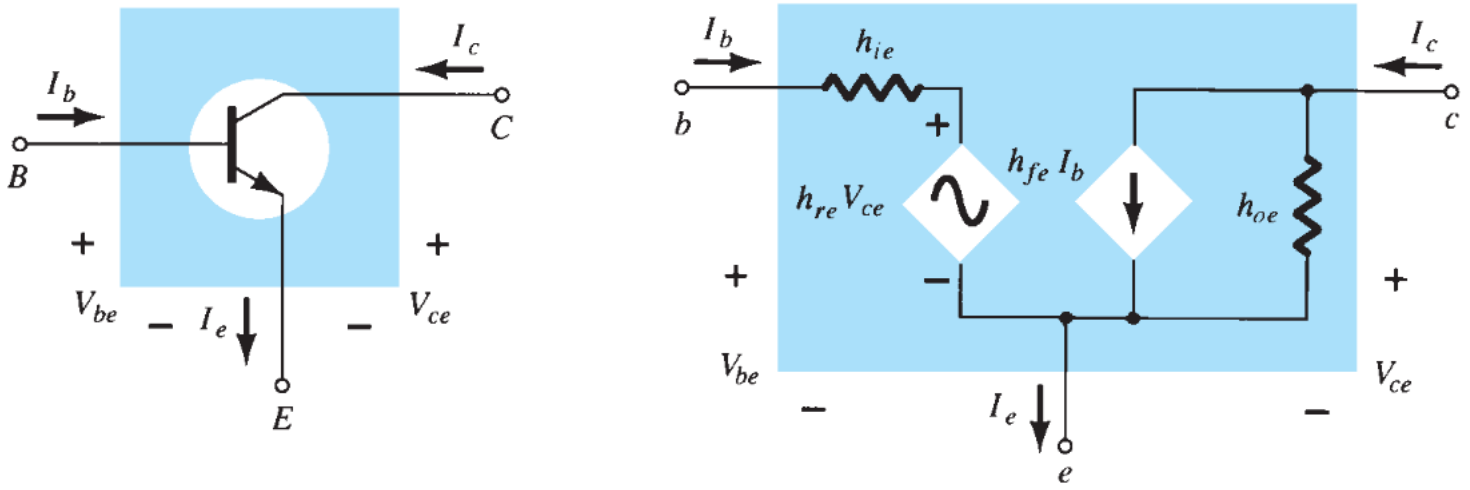
$h_{21} \rightarrow$ *f*orward transfer current ratio $\rightarrow h_f$

$h_{22} \rightarrow$ *o*utput conductance $\rightarrow h_o$



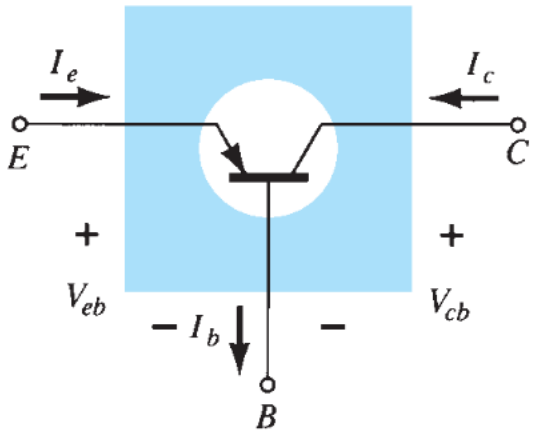
# THE HYBRID EQUIVALENT MODEL (Cont…)

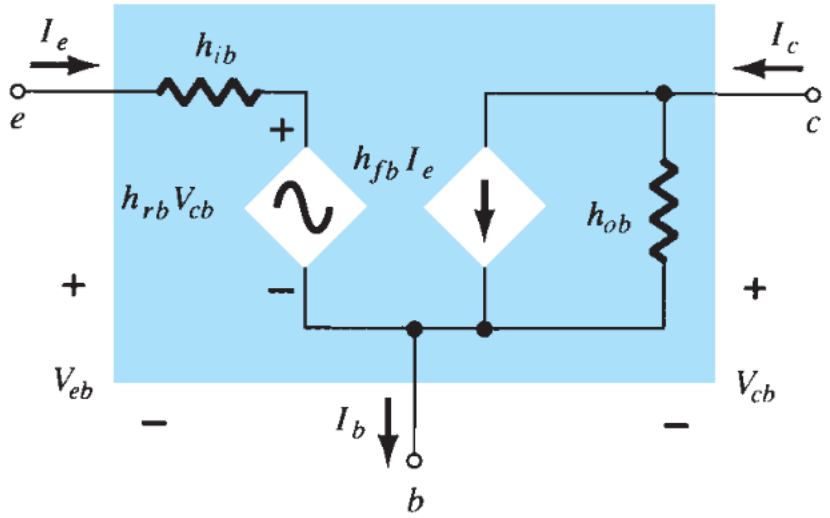
- Common-emitter configuration:



# THE HYBRID EQUIVALENT MODEL (Cont…)

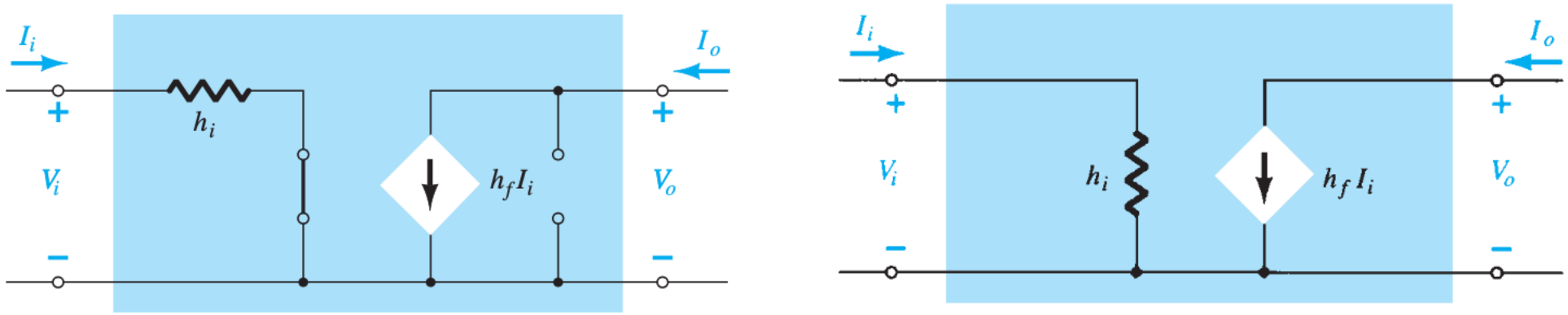
- Common-base configuration:



# THE HYBRID EQUIVALENT MODEL (Cont…)

- Because $h_r$ is normally a relatively small quality, its removal is approximated by $h_r = 0$ and $h_r V_o = 0$, resulting in a short-circuit equivalent for the feedback element.
- The resistance $1/h_o$ is often large enough to be ignored in comparison to a parallel load, permitting its replacement by an open-circuit equivalent for the CE and CB models



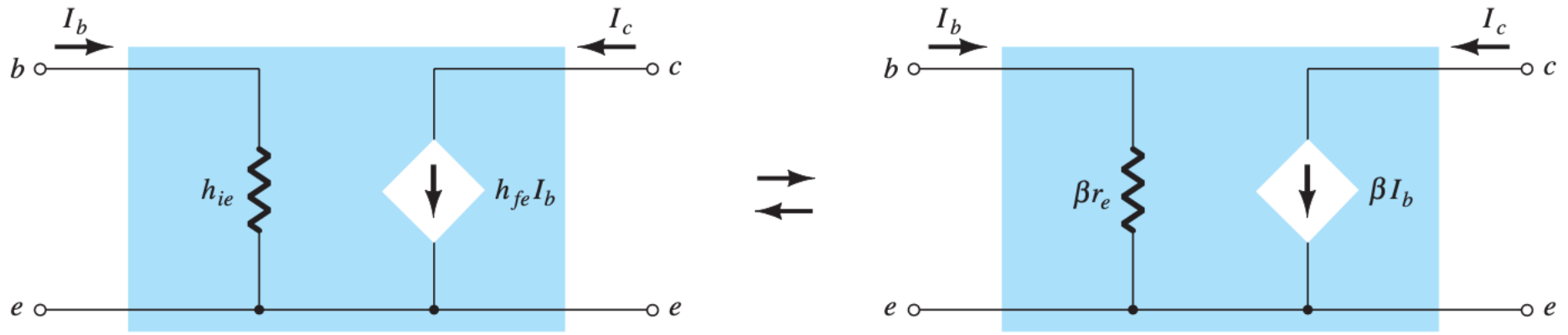
# THE HYBRID EQUIVALENT MODEL (Cont…)

## *Hybrid versus $r_e$ model*

- CE

$$h_{ie} = \beta r_e$$

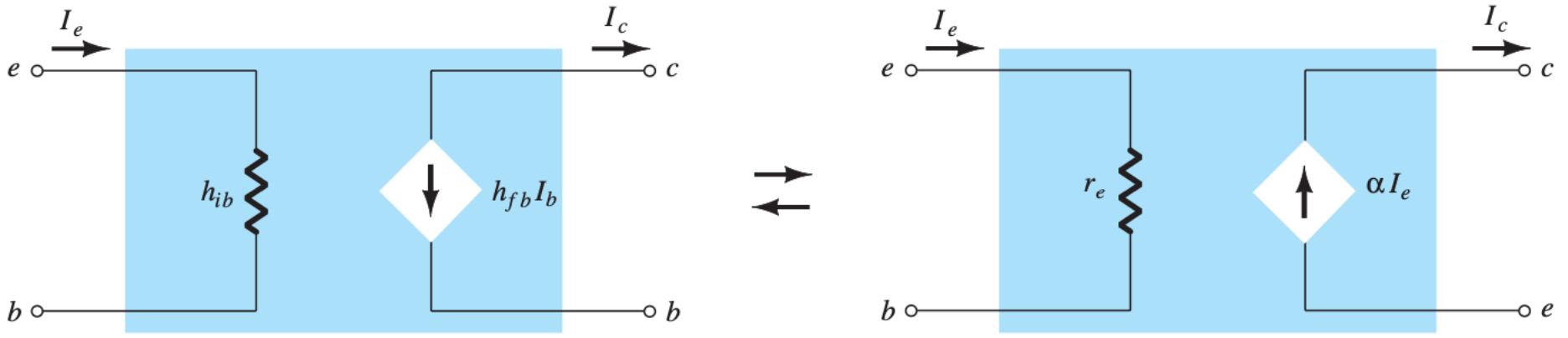
$$h_{fe} = \beta_{ac}$$




- CB

$$h_{ib} = r_e$$

$$h_{fb} = -\alpha \cong -1$$



# THE HYBRID EQUIVALENT MODEL (Cont…)

- *Hybrid parameters for the 2N4400 transistor from its datasheet*

| | | Min. | Max. | |
|---|---|---|---|---|
| Input impedance ($I_C$ = 1 mA dc, $V_{CE}$ = 10 V dc, $f$ = 1 kHz) | $h_{ie}$ | 0.5 | 7.5 | kΩ |
| Voltage feedback ratio ($I_C$ = 1 mA dc, $V_{CE}$ = 10 V dc, $f$ = 1 kHz) | $h_{re}$ | 0.1 | 8.0 | $\times 10^{-4}$ |
| Small-signal current gain ($I_C$ = 1 mA dc, $V_{CE}$ = 10 V dc, $f$ = 1 kHz) | $h_{fe}$ | 20 | 250 | — |
| Output admittance ($I_C$ = 1 mA dc, $V_{CE}$ = 10 V dc, $f$ = 1 kHz) | $h_{oe}$ | 1.0 | 30 | 1 μS |

# Introduction of Feedback Amplifiers

The phenomenon of feeding a portion of the output signal back to the input circuit is known as feedback.

The effect results in a dependence between the output and the input and an effective control can be obtained in the working of the circuit. Feedback is of two types:

- **Positive Feedback**
- **Negative Feedback**

# Introduction of Feedback Amplifiers (Cont…)

## Positive or regenerate feedback:

- In positive feedback, the feedback energy (voltage or currents), is in phase with the input signal and thus aids it. Positive feedback increases gain of the amplifier also increases distortion, noise and instability.
- Because of these disadvantages, positive feedback is seldom employed in amplifiers. But the positive feedback is used in oscillators.

# Introduction of Feedback Amplifiers (Cont...)

## Negative or Degenerate feedback:

- In negative feedback, the feedback energy (voltage or current), is out of phase with the input signal and thus opposes it.
- Negative feedback reduces gain of the amplifier. It also reduce distortion, noise and instability.
- This feedback increases bandwidth and improves input and output impedances.
- Due to these advantages, the negative feedback is frequently used in amplifiers.

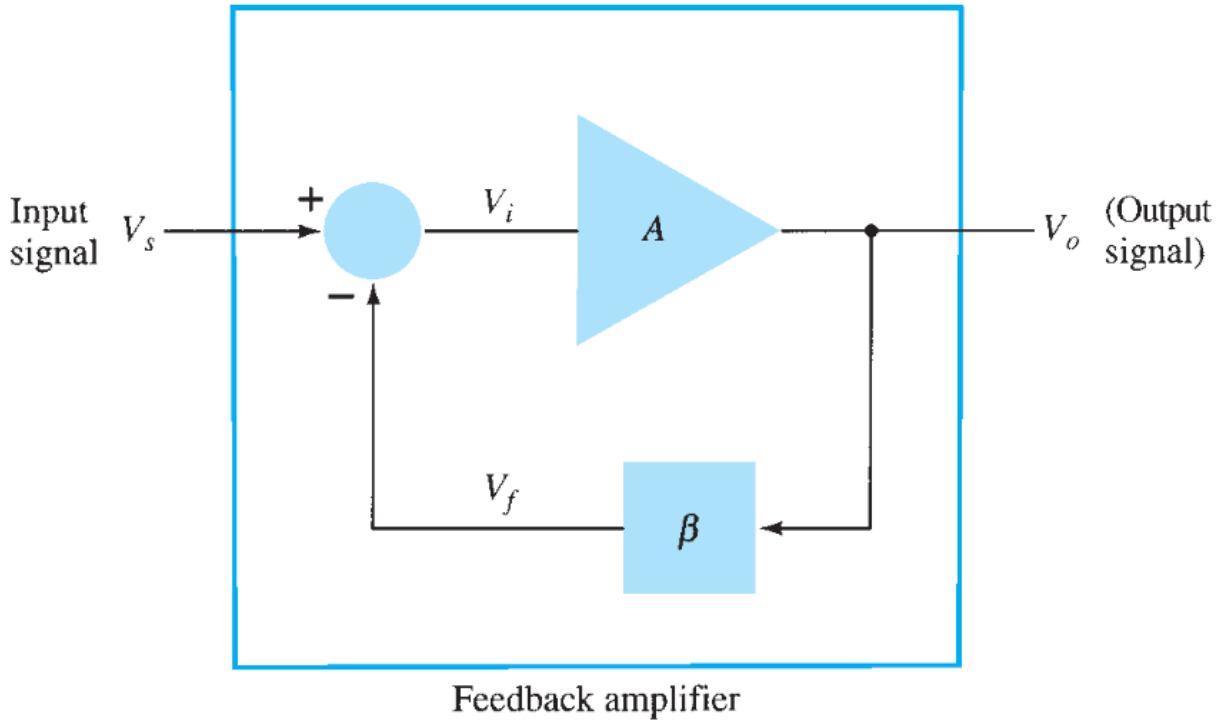
# Comparison Between Positive and Negative Feed Back

| No. | Negative Feedback | Positive Feedback |
|---|---|---|
| 1 | Feedback energy is out phase with their input signal | Feedback energy is in phase with the input signal |
| 2 | Gain of the amplifier decreases | Gain of the amplifier increases |
| 3 | Gain stability increases | Gain stability decreases |
| 4 | Noise and distortion decreases | Noise and distribution increases |
| 5 | Increase the band width | Decreases bandwidth |
| 6 | Used in amplifiers | Used in Oscillators |

# Principle of Feedback Amplifier

## *Simple block diagram of feedback amplifier*

A feedback amplifier generally consists of two parts. They are the **amplifier (A)** and the **feedback circuit (β)**. The feedback circuit usually consists of resistors.



# Principle of Feedback Amplifier (Cont…)

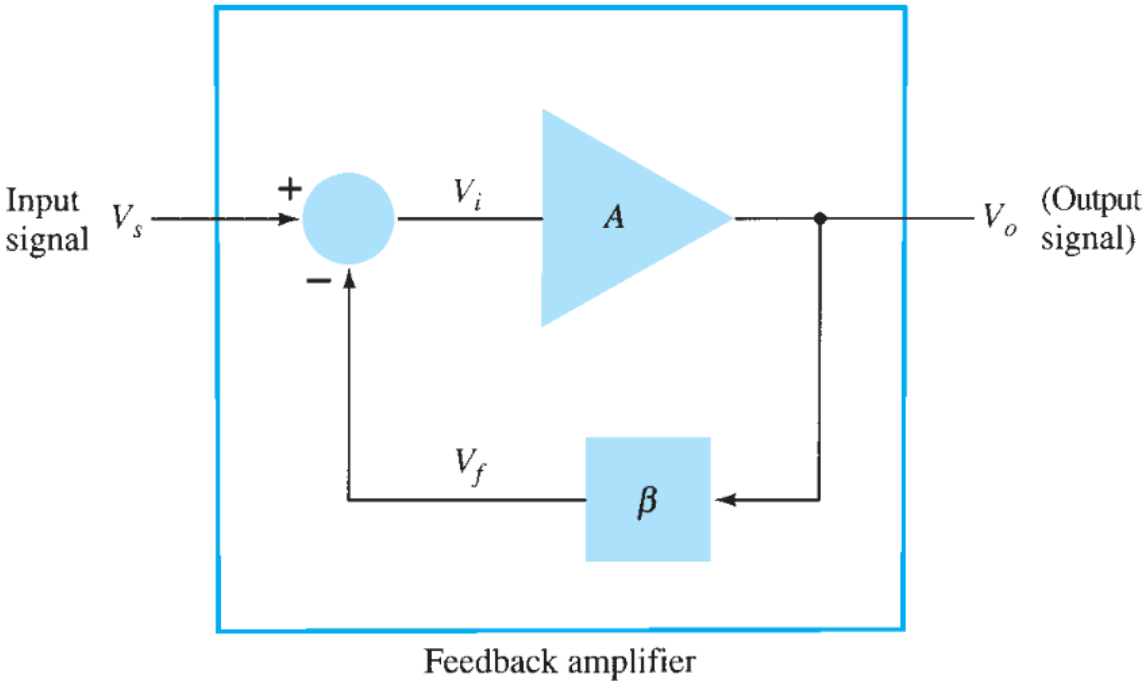
- The gain of the amplifier is represented as A. The gain of the amplifier is the ratio of output voltage $V_o$ to the input voltage $V_i$ . The feedback network extracts a voltage

**$V_f = \beta V_o$** from the output Vo of the amplifier.

- This voltage is subtracted for negative feedback, from the signal voltage $V_s$. Now,

$$V_i = V_s - V_f = V_s - \beta V_o$$

- The quantity **$\beta = V_f / V_o$** is called as feedback ratio or feedback fraction.



# Principle of Feedback Amplifier (Cont…)

- The output Vo must be equal to the input voltage (Vs - βVo ) multiplied by the gain A of the amplifier. Hence,

$$(Vs-\beta Vo)A=Vo$$

$$AVs-A\beta Vo=Vo$$

$$AVs=Vo(1+A\beta)$$

$$Vo/Vs=A/(1+A\beta)$$

- Therefore, the gain of the amplifier with feedback is given by

$$A_f=A/(1+A\beta)$$

# Feedback Connection Types

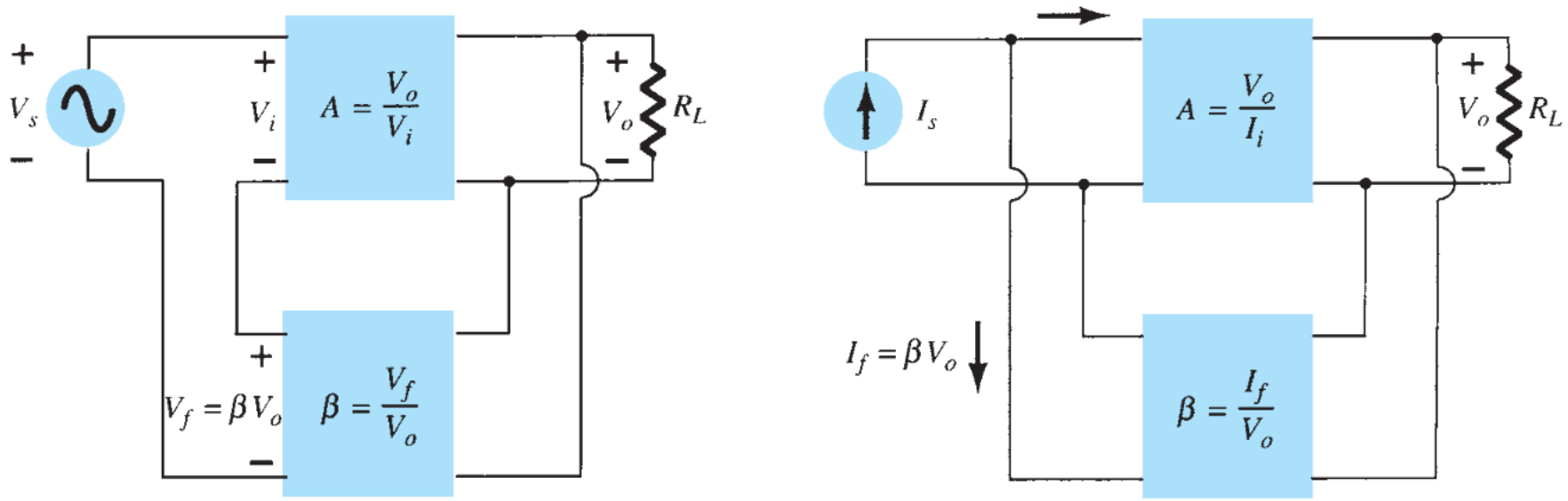
Both *voltage* and *current* can be fed back to the input either in *series* or *parallel.* Specifically, there can be:

1. Voltage-series feedback
2. Voltage-shunt feedback
3. Current-series feedback
4. Current-shunt feedback

- ***Voltage*** refers to connecting the output voltage as input to the feed-back network;
- ***Current*** refers to tapping off some output current through the feedback network;
- ***Series*** refers to connecting the feedback signal in series with the input signal voltage;
- ***Shunt*** refers to connecting the feedback signal in shunt (parallel) with an input current source.

# Feedback Connection Types (Cont…)

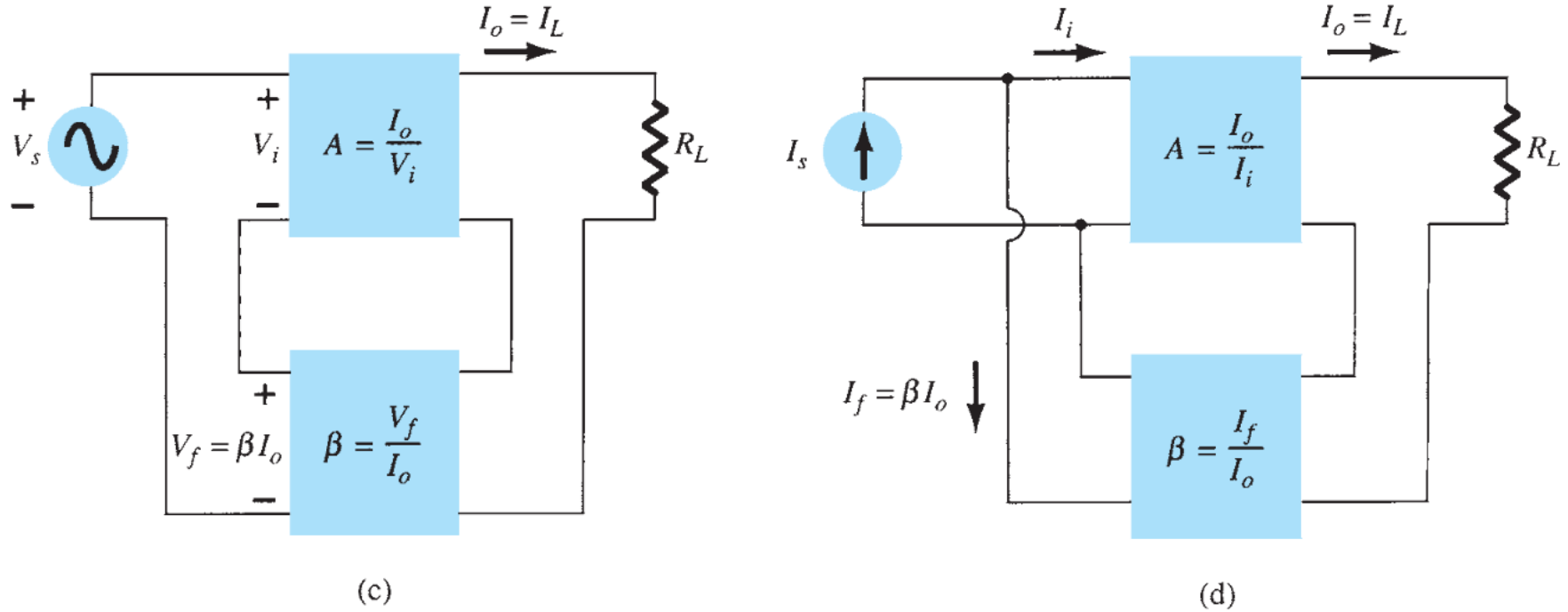
- Series feedback connections tend to *increase* the input resistance
- Shunt feed-back connections tend to *decrease* the input resistance
- Voltage feedback tends to *decrease* the output impedance




*(a) voltage-series feedback, $A_f = V_o/V_s$; (b) voltage-shunt feedback, $A_f = V_o/I_s$;*

# Feedback Connection Types (Cont…)

- Series feedback connections tend to *increase* the input resistance
- Shunt feed-back connections tend to *decrease* the input resistance
- Current feedback tends to *increase* the output impedance



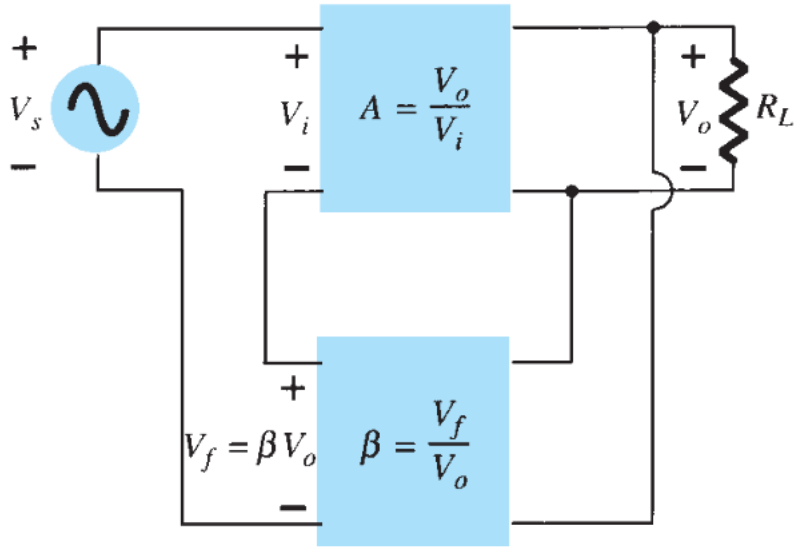

*(c) current-series feedback, $A_f = I_o/V_s$; (d) current-shunt feedback, $A_f = I_o/I_s$.*

# Gain with Feedback

The gain without feedback, *A*, is that of the amplifier stage. With feedback **β**, the overall gain of the circuit is reduced by a factor (1 + **β***A*)

| | | Voltage-Series | Voltage-Shunt | Current-Series | Current-Shunt |
|---|---|---|---|---|---|
| Gain without feedback | $A$ | $\frac{V_o}{V_i}$ | $\frac{V_o}{I_i}$ | $\frac{I_o}{V_i}$ | $\frac{I_o}{I_i}$ |
| Feedback | $\beta$ | $\frac{V_f}{V_o}$ | $\frac{I_f}{V_o}$ | $\frac{V_f}{I_o}$ | $\frac{I_f}{I_o}$ |
| Gain with feedback | $A_f$ | $\frac{V_o}{V_s}$ | $\frac{V_o}{I_s}$ | $\frac{I_o}{V_s}$ | $\frac{I_o}{I_s}$ |

# Gain as in Voltage-Series Feedback




$$A = \frac{V_o}{V_s} = \frac{V_o}{V_i}$$

If a feedback signal $V_f$ is connected in series with the input, then

$$V_i = V_s - V_f$$

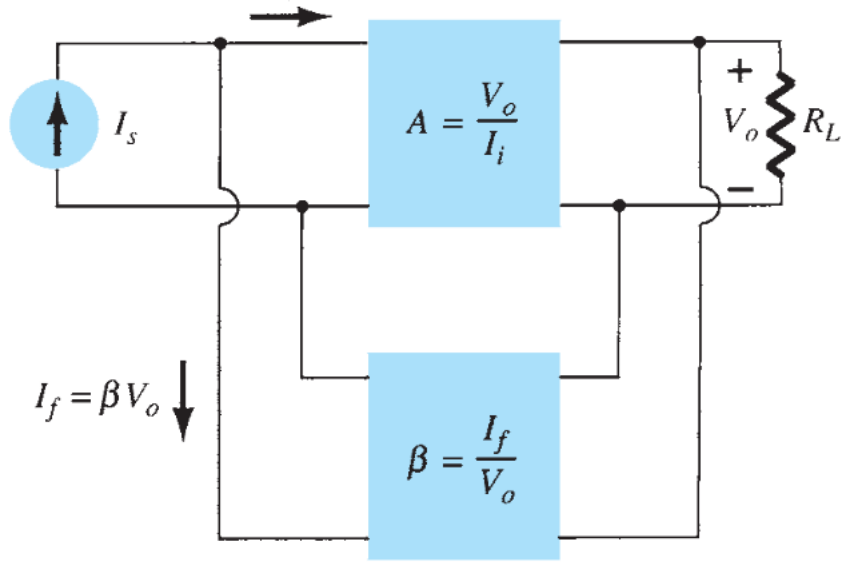
Since $$V_o = AV_i = A(V_s - V_f) = AV_s - AV_f = AV_s - A(\beta V_o)$$

then $$(1 + \beta A)V_o = AV_s$$

so that the overall voltage gain *with* feedback is

$$A_f = \frac{V_o}{V_s} = \frac{A}{1 + \beta A}$$

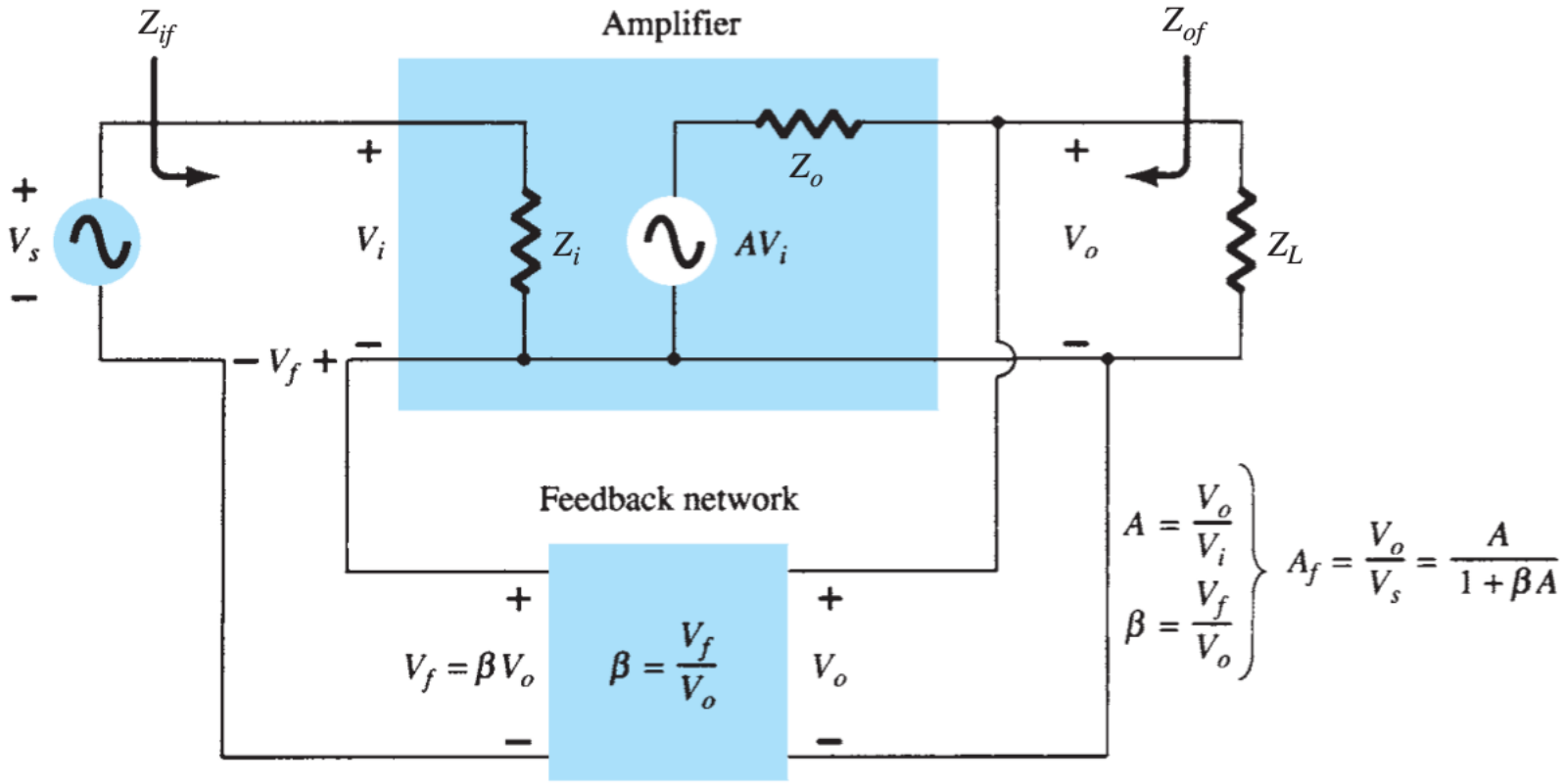
# Gain as in Voltage-Shunt Feedback




$$A_f = \frac{V_o}{I_s} = \frac{A\,I_i}{I_i + I_f} = \frac{A\,I_i}{I_i + \beta V_o} = \frac{A\,I_i}{I_i + \beta A\,I_i}$$

$$\boxed{A_f = \frac{A}{1 + \beta A}}$$

# Input Impedance of Voltage-Series Feedback



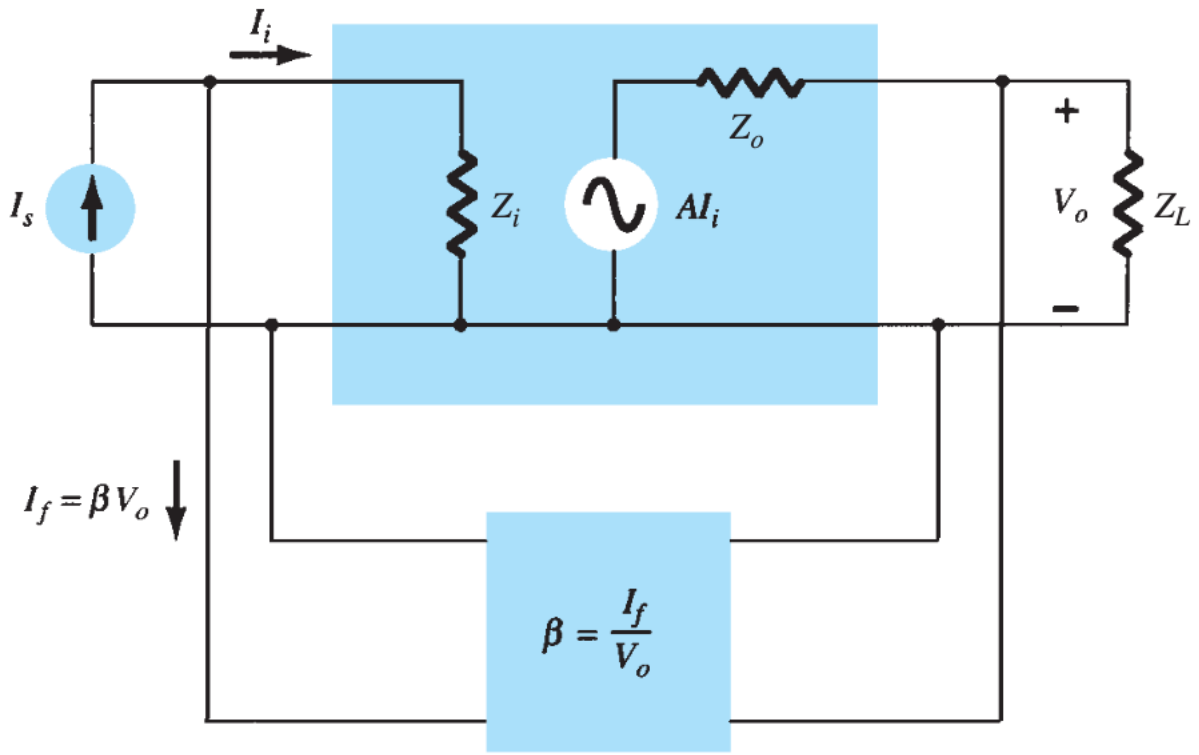

$$I_i = \frac{V_i}{Z_i} = \frac{V_s - V_f}{Z_i} = \frac{V_s - \beta V_o}{Z_i} = \frac{V_s - \beta A V_i}{Z_i}$$

$$Z_i = V_s - \beta A V_i$$

$$V_s = I_i Z_i + \beta A V_i = I_i Z_i + \beta A I_i Z_i$$

$$\boxed{Z_{if} = \frac{V_s}{I_i} = Z_i + (\beta A) Z_i = Z_i(1 + \beta A)}$$

# Input Impedance of Voltage-Shunt Feedback




$$Z_{if} = \frac{V_i}{I_s} = \frac{V_i}{I_i + I_f} = \frac{V_i}{I_i + \beta V_o}$$

$$= \frac{V_i/I_i}{I_i/I_i + \beta V_o/I_i}$$

$$\boxed{Z_{if} = \frac{Z_i}{1 + \beta A}}$$

# Output Impedance of Voltage-Series Feedback

$$V = IZ_o + AV_i$$

For $V_s = 0$, $V_i = -V_f$

so that $V = IZ_o - AV_f = IZ_o - A(\beta V)$

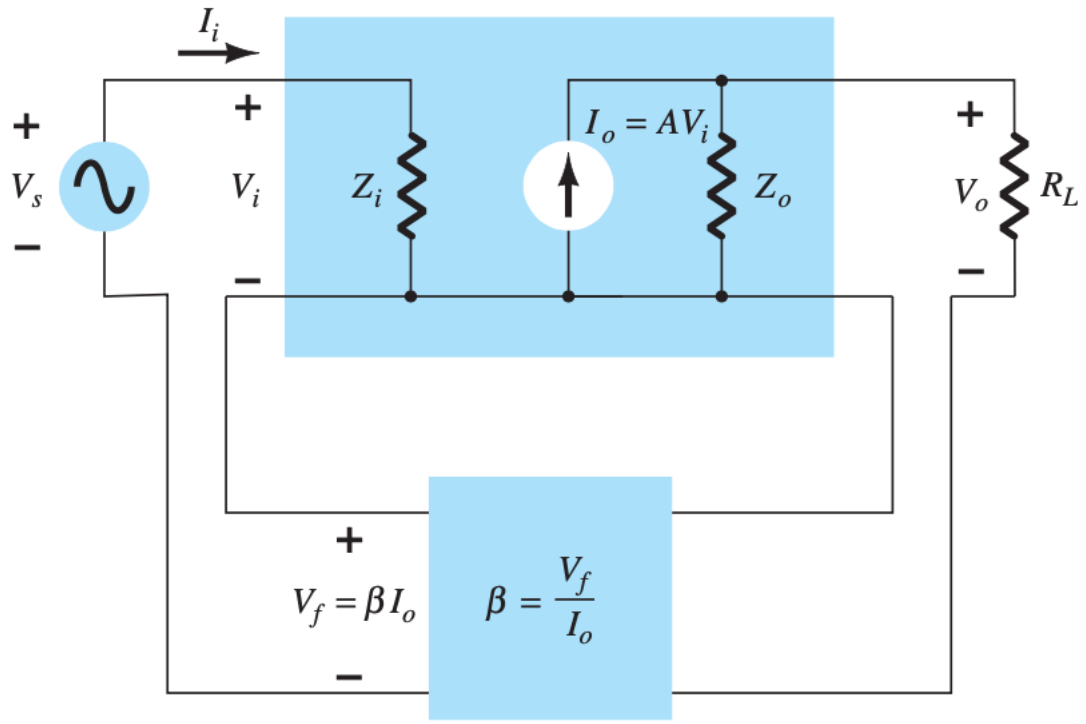
Rewriting the equation as

$$V + \beta AV = IZ_o$$

allows solving for the output impedance with feedback:

$$\boxed{Z_{of} = \frac{V}{I} = \frac{Z_o}{1 + \beta A}}$$

# Output Impedance of Current-Series Feedback




$$V_i = V_f$$

$$I = \frac{V}{Z_o} - AV_i = \frac{V}{Z_o} - AV_f = \frac{V}{Z_o} - A\beta I$$

$$Z_o(1 + \beta A)I = V$$

$$\boxed{Z_{of} = \frac{V}{I} = Z_o(1 + \beta A)}$$

# *Effect of Feedback Connection on Input and Output Impedance*

| Voltage-Series | Current-Series | Voltage-Shunt | Current-Shunt |
|---|---|---|---|
| $Z_{if}$ $Z_i(1 + \beta A)$ (increased) | $Z_i(1 + \beta A)$ (increased) | $\frac{Z_i}{1 + \beta A}$ (decreased) | $\frac{Z_i}{1 + \beta A}$ (decreased) |
| $Z_{of}$ $\frac{Z_o}{1 + \beta A}$ (decreased) | $Z_o(1 + \beta A)$ (increased) | $\frac{Z_o}{1 + \beta A}$ (decreased) | $Z_o(1 + \beta A)$ (increased) |